

TelegramMQL5

TelegramMQL5 is a Python project that listens to a Telegram group, parses trading signals, and saves them in JSON format for integration with MQL5 trading platforms.

It supports **MessageFilter parsing**, **signal mapping**, and **secure login** via Telegram API with QR code or 2FA.

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Features

- Listen to a Telegram group for trading signals.
 - Parse messages using `MessageFilter` and handle errors gracefully.
 - Supports multi-line signals with emojis and multiple TP/SL values.
 - Map user-friendly names (`GOLD`, `SILVER`) to actual trading symbols (`XAUUSD`, `XAGUSD`).
 - Save signals in `signals.json` with timestamps and metadata.
 - Easy manual testing via `test.py`.
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Requirements

- Python 3.11+

Install python from this [python official page](#)

Installation

1. Download .rar given.
2. UnZip it

3. Open **cmd** window : Alt + Window => cmd
4. Go to folder TelegramMQL5

```
cd /d C:\path\to\unziped\TelegramMQL5
```

5. Create a virtual environment (optional but recommended):

```
python -m venv venv  
venv\Scripts\activate
```

6. Install requirements:

```
pip install -r requirements.txt
```

```
C:\Users\User>cd /d D:\Desk\python\TelegramMQL5  
D:\Desk\python\TelegramMQL5>python -m venv venv  
D:\Desk\python\TelegramMQL5>  
D:\Desk\python\TelegramMQL5>venv\Scripts\activate  
(venv) D:\Desk\python\TelegramMQL5>pip install -r requirements.txt  
Collecting colorama==0.4.6 (from -r requirements.txt (line 1))  
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)  
Collecting dotenv==0.9.9 (from -r requirements.txt (line 2))  
  Downloading dotenv-0.9.9-py2.py3-none-any.whl.metadata (279 bytes)  
Collecting lang==0.1 (from -r requirements.txt (line 3))  
  Downloading lang-0.1-py3-none-any.whl.metadata (1.7 kB)  
Installing collected packages: telegram, pyaes, pyyaml, python-dotenv, pyasn1, lang, colorama, rsa, qrcode  
Successfully installed PyYAML-6.0.3 Telethon-1.42.0 colorama-0.4.6 dotenv-0.9.9 lang-0.1 pyaes-1.6.1 pyasn1-0.5.1 qrcode-8.2 rsa-4.9.1 telegram-0.0.1  
[notice] A new release of pip is available: 24.0 -> 25.3  
[notice] To update, run: python.exe -m pip install --upgrade pip  
(venv) D:\Desk\python\TelegramMQL5>
```

Now you have all environnement set.

Configuration

Go to the provided config template **TelegramMQL5/app/config/config.ini** and fill in your credentials:

```

[telegram]
api_id = 123456
api_hash = abcdef1234567890
session_name = telegram_session
group_id = -12345678912

[paths]
mql_file_dir = C:\Users\User\AppData\Roaming\MetaQuotes\Terminal\
<YourTerminalID>\MQL5\Files

[trading]
pairs = XAUUSD, EURUSD, SILVER, BTCUSD, GOLD

[pair_mapping]
GOLD = XAUUSD
SILVER = XAGUSD
ETH = ETHUSD

[mql]
default_lot = 0.01
comment = TelegramSignal

```

⚠ Important – Required Configuration Variables

Before running the bot, you must carefully fill in the following variables in config.ini. Incorrect values will prevent TelegramMQL5 from connecting to Telegram or saving signals correctly.

- **api_id / api_hash**: obtained from my.telegram.org.
 - Tutorial [here](#) at 2m35
- **session_name**: the filename used to store your session (`telegram_session.session`).
- **group_id**: ID of the Telegram group you want to get messages from.
 - **Required**: If this is incorrect, the bot won't receive messages.
 - The ID can be found with command : "python -m app.get_id_channels"
- **mql_file_dir**: Replace and with your Windows username and your MT5 terminal ID..
 - **Required**: This ensures that the bot writes the signals JSON (signals.json) into the correct folder that your EA will read.
- **pairs**: list of trading symbols the parser will accept.
- **pair_mapping**: maps user-friendly names used in Telegram messages to the actual trading symbols on MQL5.
 - For example, if the Telegram message contains **GOLD**, **pair_mapping** allows the parser to automatically convert it to **XAUUSD**, which is the symbol used in your MQL5 platform.
 - This ensures that messages like **Buy: GOLD** or **Sell: SILVER** can be correctly interpreted and saved with the proper symbol for trading.
- **default_lot**: the lot you want to open in your trade.

Usage

Running the main bot

```
python -m app.main
```

How it works:

- The bot will automatically check if a Telegram session file (`session_name.session`) exists.
- **If the session does not exist** (first run):
 1. A **QR code** will appear in the terminal.
 2. Scan the QR code with your Telegram app (Desktop or Mobile).
 3. The login is authorized and a new session file is created automatically.
- **If your account has 2FA enabled**, the bot will prompt you to enter your password after scanning the QR code.
- On subsequent runs, the bot will **reuse the existing session file** automatically, so you won't need to scan the QR code again.
- Once connected, the bot will start listening to the configured Telegram group and log all messages in `logs/telegram_log.txt`.

Example workflow for the first connection:

1. Run the bot:

```
python -m app.main
```

2. Terminal displays a QR code:



3. Open your Telegram app → Settings → Devices → Scan QR code
4. Telegram authorizes the session and creates `telegram_session.session`
5. Bot prints:

```
✓ Logged in as: (username)  
Listening...
```

```
(venv) D:\Desk\python\TelegramMQL5>python -m app.main
No existing session found, performing login...
Scan this QR code with your Telegram app:
```



```
👤 Logged in as: @user (TelegramMQL5)
Listening...
```

From now on, you can just run the bot without scanning the QR code again.

6. To stop the bot: press **CTRL + C**. Terminal will show:

```
KeyboardInterrupt
Disconnected from Telegram.
```

Getting Group ID

To listen to a **Telegram group**, the bot need:

- **group_id** → for private groups (numeric ID, starts with **-100...**)

For **private groups**, you cannot use the name. You must use the numeric ID.

Step 1 – Run the ID retrieval script

TelegramMQL5 includes a script that lists all the groups your account belongs to, along with their IDs:

```
python -m app.get_id_channels
```

Step 2 – Read the output

The script will display something like this:

```
MyPublicGroup -100987654321
TradingSignalsPrivate -1001234567890
SomeOtherGroup -1001122334455
```

- **Name** → name of the group as it appears in Telegram
- **ID** → numeric group ID (use this for private groups)

Step 3 – Update `config.ini`

Copy the numeric ID of your group and paste it in the config:

```
[telegram]
group_id = -1001234567890 ; Private group ID
```

Step 4 – Run the bot

```
python -m app.main
```

- The bot will listen to the group using the **ID for private groups**.
- This avoids errors like `ValueError: Cannot find any entity corresponding to "MyPrivateGroup"`.

Testing MessageFilter

```
python -m app.test
```

- Allows you to paste **any Telegram message** manually.
- Multi-line messages with emojis are supported.

Example message to test:

```
Buy : XAUUSD
🎯 Entry : 4822
⊖ Stop : 4812,62
🚀 TP 1 : 4842,6
```

- Paste this message in the terminal.
 - The parser will detect **direction**, **entry price**, **stop loss**, and all **TP values**.
 - Any parsing errors are logged using `FailedParseMessage`.
-

Session Management

- Session files are saved as `session_name.session`.
- Telethon will automatically reuse existing sessions.
- To force a new login, delete the session file:

```
del telegram_session.session # Windows
```

- If no session exists, a **QR code** will appear in the terminal for login.
 - If your account has 2FA enabled, the script will prompt for the password.
-

MQL5 Integration

To use the signals parsed by **TelegramMQL5** in MetaTrader 5, follow these steps:

1. Copy EA files

- Copy the EA file from the repository:

```
TelegramMQL5/TelegramToMQL5.ex5
```

- Paste it into your MetaTrader 5 Experts folder:

```
C:\Users\<YourUser>\AppData\Roaming\MetaQuotes\Terminal\  
<YourTerminalID>\MQL5\Experts
```

Replace `<YourUser>` and `<YourTerminalID>` with your Windows username and your MT5 terminal ID.

2. Launch MetaTrader 5

- Open MT5 and load the Expert Advisor (EA) on **any chart**.
- In the **Experts** window, verify that there are **no errors**.
- The EA will automatically read the signals from `signals.json` and execute trades according to your settings.

4. Run TelegramMQL5

- Start the Python bot as usual:

```
python -m app.main
```

- Wait for messages in your configured Telegram group.
- Each parsed signal is saved in `signals.json` inside the MT5 `Files` folder.
- The EA will pick up these signals and execute trades accordingly.

⚠ Make sure your `pair_mapping` in `config.ini` correctly maps the names in Telegram messages (e.g., `GOLD`) to the MQL5 symbols (e.g., `XAUUSD`). This ensures that the EA executes trades on the correct instrument.

The screenshot shows a Telegram chat interface at the top with a message: "Buy GOLD at 4931 SL at 4886.5 TP at 4955 14:52". Below the chat is the MT5 Experts window. A red arrow points to the message in the chat with the text "Receive new message on Telegram". Another red arrow points to the message in the Experts window with the text "Log in Experts window". A third red arrow points to the "Opening Trade" button in the Experts window with the text "Opening Trade". The Experts window displays a table of trade data:

Symbol	Ticket	Time	Type	Volume	Price	S / L	T / P	Price	Swap
xauusd	1425399822	2026.01.23 15:52:32	buy	0.01	4936.83	4886.50 ✕	4955.00 ✕	4936.43	

At the bottom of the Experts window, the account balance is shown: "Balance: 4 980 66 USD, Equity: 4 980 26, Margin: 9.87, Free Margin: 4 970 39, Margin Level: 50.458.56 %".

License

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